Sentinel

A Privacy-Preserving Edge-AI Architecture for Proactive Mental Health Anomaly Detection

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Abstract

More than one billion people currently live with a diagnosable mental health condition, and suicide remains among the leading causes of death in adolescents and young adults worldwide. Passive smartphone sensing — termed digital phenotyping — has been proposed as a scalable complement to episodic clinical assessment, since continuously generated behavioural signals (communication patterns, circadian rhythm, mobility, application usage) can reveal symptom trajectories between clinical visits. However, existing digital-phenotyping research infrastructure and consumer wellness features rely on cloud-mediated processing or centralized data aggregation, creating a structural conflict between clinical utility and user privacy that has limited real-world, at-scale deployment. This paper presents Sentinel, an architecture for proactive mental health anomaly detection, proposed as a standalone Android application requesting the same class of OS-level behavioural permissions (`PACKAGE_USAGE_STATS`, `ACTIVITY_RECOGNITION`) already used by existing digital-wellbeing and screen-time applications — a deployment path with direct commercial precedent, and one that a first-party integration into Android's own Digital Wellbeing suite could extend but does not require — under a strict zero-knowledge constraint: no raw behavioural data — text, media, location, or usage logs — ever leaves the device. Sentinel couples a two-phase on-device TensorFlow Lite pipeline (lightweight NLP sentiment/risk scoring via TF-IDF and MobileBERT variants, Gemma 4, followed by an ensemble anomaly-detection layer wrapped in conformal calibration) with data-at-rest encryption via SQLCipher and hardware-backed key storage in the Android Trusted Execution Environment. Generation of active, conversational content is grounded in clinical instruments via a retrieval-augmented mechanism and verified by a dedicated safeguarding layer before reaching the user, and a turn-level conversational classifier — adapted from a recently published clinical guardrail model — screens the user's own replies during these check-ins for self-harm or harm-to-others risk. Critically, Sentinel replaces automated third-party notification with a graduated, human-anchored intervention protocol, escalating to existing human-staffed crisis infrastructure by default, with any external contact reserved for a user-preconfigured, explicitly opted-in mechanism. We describe the system architecture, examine the computational feasibility of continuous on-device inference under mobile hardware constraints, propose a staged validation plan extending existing evaluation frameworks to the detection side of the system, and outline an ethical framework addressing false-positive harm, validation without clinical ground truth, cultural and interpretive robustness, and the risk of intervention-protocol misuse.

1. Introduction

Mental health disorders now represent one of the most pressing challenges in global public health. World Health Organization estimates place over one billion people worldwide as living with a mental health condition, with anxiety and depressive disorders the most prevalent across both sexes. Suicide alone accounted for an estimated 727,000 deaths in 2021 and remains the third leading cause of death among 15–29 year-olds globally [2]. Yet clinical detection of deteriorating mental state continues to rely overwhelmingly on episodic, self-reported assessment — structured interviews and standardized questionnaires administered at intervals of weeks or months — a cadence poorly matched to the temporal dynamics of mood and suicidal ideation, which can fluctuate meaningfully within a single day. This gap has not gone unnoticed by the population it affects most: a 2026 nationally representative survey of US adolescents and young adults found that nearly one in five had used an AI chatbot for mental health advice, and that 63% of those users had never disclosed that use to a parent, clinician, or other trusted adult [13] — a pattern that speaks as much to unmet need as it does to the privacy stakes of any system built to observe this population's behaviour.

Smartphones, carried continuously and instrumented with an increasingly rich sensor suite, offer a candidate solution. The resulting research paradigm — digital phenotyping, defined as the moment-by-moment quantification of individual-level human behavior using data from personal digital devices [1] — has motivated over a decade of work correlating passively sensed behavioral markers with depressive symptom severity. Saeb et al. found that GPS-derived features (variance in locations visited, regularity of 24-hour movement rhythm) and phone-usage frequency correlated significantly with depression severity over a two-week sensing period across 40 adults [3]. The StudentLife study similarly found that accelerometer-derived stationary time and screen-unlock duration distinguished depressed from non-depressed students across a full academic term, in a cohort of 83 participants [4]. Open research platforms such as Beiwe, developed at the Harvard T.H. Chan School of Public Health [1], and mindLAMP, developed at Beth Israel Deaconess Medical Center / Harvard Medical School [5], have operationalized this paradigm at scale, coupling passive sensor collection with clinician-facing dashboards for research and, increasingly, clinical use.

This clinical promise has not translated into consumer-scale deployment, largely because existing digital-phenotyping infrastructure was built for research data collection rather than end-user privacy. Beiwe, mindLAMP, and comparable platforms are explicitly designed to transmit sensor and survey data to a server for aggregation and clinician review — an appropriate model for IRB-governed research, but a poor fit for an OS-level consumer feature where the user, not a research team, must remain the sole party with access to their own behavioral data. The broader machine learning community has converged on a partial answer to this tension: federated and on-device learning architectures, which keep raw data on the client device and transmit only model updates or, in the fully local case, nothing at all [6]. What remains largely unaddressed is combining this privacy posture with the specific, high-stakes task of detecting acute mental health risk, where the cost of a missed detection and the cost of a false positive are both severe and asymmetric.

Intervention design compounds this tension. Prior systems for detecting self-harm and suicide risk in user-generated text — including recent work applying transformer-based classifiers to private social-media conversations, built specifically to distinguish casual expression from genuine risk [7] — have generally been built for platform-side content moderation, where a detected signal triggers a report to human moderators. This design assumes a trust relationship between the detecting system and any party it notifies; that assumption does not hold for an OS-level, self-monitoring feature, where a poorly designed escalation path can itself become a vector for harm. This is not a hypothetical concern: a substantial body of HCI security research documents how ostensibly protective device features — shared location, shared passwords, "for your safety" monitoring — are routinely repurposed by abusive partners for coercive control [8]. A pre-authorized emergency contact is exactly this shape of feature. To our knowledge, no existing system combines (i) fully local, multimodal behavioral anomaly detection built on standard, OS-level behavioural permissions rather than a research-grade data pipeline, (ii) hardware-backed, zero-knowledge data-at-rest encryption, and (iii) a graduated, human-in-the-loop intervention protocol that defaults to user-directed crisis resourcing rather than automated third-party notification. Section 2.1 examines a cluster of recent edge-LLM mental health systems that share pieces of this ambition, and the specific gaps in each that this claim rests on. This is the gap Sentinel is designed to address.

This paper makes three contributions. First, we present the Sentinel system architecture: a two-phase, on-device TensorFlow Lite pipeline for behavioral feature extraction and unsupervised anomaly detection, integrated with Android's native usage, activity-recognition, and Trusted Execution Environment APIs, together with a retrieval- and instrument-grounded generation layer for active conversational check-ins and a turn-level risk classifier screening the user's replies within them. Second, we examine the computational feasibility of this architecture under real mobile hardware constraints, including the specific limitations of on-device vision models for the more exploratory task of detecting self-harm-related themes in user-generated media. Third, we propose an ethical framework and staged validation plan for the intervention protocol, addressing false-positive harm mitigation, user autonomy, interpretive and cultural robustness, and the risk of protocol misuse in coercive-control contexts. Section 2 reviews related work across digital phenotyping, conversational agents, generation safety, and prediction under uncertainty; Section 3 details the proposed methodology; Section 4 examines implementation feasibility; Section 5 develops the ethical framework; and Section 6 concludes with directions for future validation work.

2. Literature Review

Sentinel sits at the intersection of six research threads: digital phenotyping and passive sensing, conversational agents in mental health, the emerging empathy-paradox literature on what these systems actually understand, generation-safety techniques including retrieval-augmented generation, the statistics of predicting rare events, and the regulatory and ethical scaffolding now forming around clinical AI. This section reviews each in turn and closes by positioning Sentinel against the gap they collectively leave open.

2.1 Digital Phenotyping: From Passive Sensing to Behavioral Prediction

The term digital phenotyping was introduced by Torous, Kiang, Lorme, and Onnela [1], who in the same paper released Beiwe, an open research platform for passive smartphone sensing. Saeb et al. [3] gave this premise its first strong empirical footing, showing that GPS-derived and phone-usage features correlated significantly with depression severity. The StudentLife project [4] extended this to a full academic term, finding that accelerometer-derived stationary time and screen-unlock duration distinguished depressed from non-depressed participants. Both studies converge on the finding that motivates Sentinel's own behavioral feature set: sleep, mobility, and app-engagement signals are the strongest and most consistently replicated correlates in this literature — stronger than anything derived from media content.

Beiwe's later counterpart, mindLAMP [5], operationalized this at greater scale by pairing passive sensing with brief active check-ins (Ecological Momentary Assessment) and clinician-facing dashboards. Both platforms share a structural feature that matters for Sentinel's design: they transmit sensor and survey data to a server for clinician review, appropriate for IRB-governed research but a poor fit for a consumer-facing, zero-knowledge feature. The broader machine learning field has partially answered this tension through federated and on-device learning, formalized by McMahan et al. [6], which keeps raw data on the client device and transmits only model updates, or nothing at all in the fully local case — the case Sentinel adopts as a hard constraint.

One further signal resolves a tension in Sentinel's own privacy posture: keystroke dynamics. The BiAffect research program [20, 21] demonstrated, across a bipolar cohort and a large open-science iOS sample, that typing metadata alone — inter-key timing, backspace rate, session duration and variability — correlates with clinician-administered mood severity, without ever needing to process the content of what was typed. This is a stricter privacy standard than Sentinel's text-sentiment layer and is incorporated into the feature set for that reason; Section 3.2 details how Sentinel's own capture surface for this signal differs from BiAffect's original delivery mechanism.

Recent edge-LLM systems for mental health. A distinct and faster-moving thread applies the same on-device philosophy not to passive behavioral sensing but to conversational or diagnostic LLM inference directly. Three recent systems are close enough to Sentinel's framing that a reviewer familiar with the space would expect them addressed directly, and each leaves open a specific gap that motivates a design choice in Sentinel rather than simply competing with it.

Ji et al.'s MindGuard [57] is the closest in ambition: an edge LLM that fuses objective mobile-sensor data with subjective Ecological Momentary Assessment records to deliver screening and intervention conversations, evaluated on open datasets spanning four years and validated through a real-world deployment across mobile devices involving 20 subjects over two weeks, with reported performance comparable to GPT-4 despite a far smaller model. That validation is real but small — a 20-participant, two-week pilot is evidence of feasibility, not of sustained adoption — and, more consequentially for Sentinel's own design, the system's sensor pipeline depends on an external Fitbit for physiological signal, introducing a second piece of hardware, a pairing step, and a charging habit as adoption prerequisites that a phone-only architecture avoids entirely.

Bandara et al.'s zero-egress psychiatric AI platform [56] pursues the same on-device, zero-egress privacy posture Sentinel adopts, but for a categorically different task: an ensemble of three fine-tuned LLMs (Gemma, Phi-3.5-mini, Qwen2) producing DSM-5-aligned differential diagnoses from clinician- or patient-entered conversational text, aimed primarily at military, correctional, and remote clinical settings rather than consumer self-monitoring. The system is entirely conversation-triggered — it assesses only what a clinician or patient actively types into a diagnostic interview — and its own reported evaluation is explicit about its limits: fine-tuning used only 500 synthetic conversational records, and the paper's latency figures are described by its authors as single-session, uncontrolled, demo-observed estimates, with a controlled benchmark study and a prospective clinical validation named as future work rather than completed. A production clinical deployment is described as planned, in collaboration with a named U.S. Army health center, not yet realized.

MoPHES [58] fine-tunes two compact (0.5B-parameter) on-device models — one for mental-state severity assessment, one for empathetic dialogue — and, like Bandara et al.'s system, assesses mental state exclusively from what a user actively types to the agent, re-evaluating severity every five conversational turns; it has no passive-sensing component at all. Its training and evaluation data is synthetic throughout: single-turn, web-sourced counselling Q&A pairs, expanded into multi-turn dialogues by prompting GPT-4o-mini, with performance reported against automated metrics (BLEU, ROUGE) and small-scale manual scoring rather than any real-user or field deployment.

All three systems, in other words, share the property that motivates one of Sentinel's central design choices: each requires the user (or a clinician) to already recognize something is wrong and initiate an explicit diagnostic or therapeutic conversation before the system can assess anything. None has been validated at a scale or duration — a 20-subject, two-week pilot at the high end; a single uncontrolled demo session at the low end — that would constitute evidence of real-world adoption rather than research-stage feasibility. Sentinel's passive, ambient detection layer (Section 3.2) is designed to act as a complement to exactly this class of system, surfacing a check-in prompt before the user necessarily has the words — or the self-recognition — to start one themselves; Section 3.6 describes how one of these systems, MindGuard [57], nonetheless directly informs a component of Sentinel's own conversational safety design, once the specific conversational surface it needs is scoped correctly.

2.2 Conversational Agents in Mental Health Care

The idea that a machine could deliver therapeutic conversation predates modern NLP by half a century: Weizenbaum's ELIZA (1966) simulated a Rogerian therapist through simple pattern matching, and users nonetheless disclosed intimate material to it — an early instance of what human-computer interaction research later formalized as the Computers Are Social Actors paradigm. Modern systems have made this dynamic clinically actionable. Woebot, evaluated in a randomized controlled trial by Fitzpatrick, Darcy, and Vierhile [17], delivered CBT to young adults with depression and anxiety symptoms with measurable symptom reduction. Wysa, evaluated by Inkster et al. and summarized in Dehbozorgi et al.'s 2025 systematic review [49], showed that among self-reporting users, higher engagement predicted significantly larger symptom improvement, with 67.7% of users rating the app helpful — an early demonstration that engagement depth, not mere access, drives outcome, the logic behind Sentinel's graduated tiers rather than a single flat intervention.

Not every result in this literature is encouraging in the way marketing materials suggest, and the honest ones are more useful to Sentinel than the flattering ones. Tate et al., also reviewed in Dehbozorgi et al. [49], built a machine learning model to predict adolescent mental health problems from a large Swedish twin cohort and achieved their best result — AUC 0.739 with a random forest — concluding explicitly that the model, despite being the strongest configuration tested, was not accurate enough for clinical use. This is a rare and valuable admission in a literature that Dehbozorgi et al.'s own review characterizes as frequently suffering from inadequate transparency in reporting model features and an overly optimistic view of AI performance — a field-level critique this paper takes as a standard to write against.

Kellogg and Sadeh-Sharvit [53] offer a complementary, practice-oriented lens: rather than asking whether AI can replace clinicians, they organize mental-health AI into three categories clinicians actually interact with — automation, engagement, and clinical decision-support technologies — and document the real friction of integrating each into daily practice. This framework maps cleanly onto Sentinel's own architecture and is used in Section 3 to justify why the human-counsellor handoff needs its own onboarding, not only user-facing design.

2.3 The Empathy Paradox and the Boundary Between Expression and Interpretation

Lawrence et al.'s survey of the opportunities and risks of large language models in mental health [30] frames the tension this section examines at the level of the whole field: the conversational fluency that makes these systems accessible and engaging is the same property that makes their failures hard to detect, since a response can be linguistically appropriate while being clinically wrong. Sentinel's own generation layer inherits this exact tension, and the remainder of this section examines it in more specific, technical form.

A 2026 study by Kehinde et al. [10] extracted 1,000 AI-directed and 1,000 human-therapist-directed mental health conversations and used a Random Forest classifier to distinguish them by emotional content across five domains. The classifier achieved 48.3% accuracy — chance level — with the modest features that did distinguish the groups being structural (word count) rather than emotional. Users, in other words, express distress to AI systems with the same linguistic and emotional texture used with human clinicians. This finding is easy to over-read: it establishes that human expression toward AI is genuine, not that AI interpretation of that expression is reliable. These are different claims, and the literature on the second is considerably more cautionary.

The Thera-Turing Test [12] makes this distinction structurally explicit: its evaluation framework separates "therapeutic relationship skills" (is the agent empathic, rapport-building) from "treatment fidelity" and "safety" (is the agent's underlying clinical judgment correct), and none of its four evaluation methods specifically stress-tests interpretation under deliberately ambiguous or adversarial input. That gap is not hypothetical. EmoAgent [11], comprising EmoEval and EmoGuard, simulated vulnerable users interacting with unguarded conversational agents and found psychological deterioration in more than a third of interactions, with delusion-exacerbation rates exceeding 90% for certain agent personas — a documented case of a system sounding appropriately warm while actively worsening the underlying clinical picture. A parallel literature on AI-associated sycophancy makes the same point with more clinical weight: a documented industry rollback of a large conversational model update specifically for validating user doubts and reinforcing negative emotion, and the "Technological folie à deux" analysis of chatbot–mental-illness feedback loops [31], both document sycophantic validation as an active harm mechanism, not a benign stylistic quirk.

A second, distinct axis of interpretive risk is cultural and linguistic. Kleinman's foundational work on idioms of distress [32], extended by Kohrt and Hruschka's demonstration that expressions such as the Nepali man dukheko do not map cleanly onto Western depression constructs [33], establishes that culturally coded expressions of distress can be systematically misread by systems calibrated to Western/English norms. This concern is current and active: PsychiatryBench [34] was built explicitly to test whether large language models apply ethnocentric assumptions when interpreting culturally mediated clinical language, and a companion effort on cross-lingual mental health ontologies for Indian languages [35] addresses the same problem in the linguistic context most relevant to Sentinel's integration with Tele

MANAS [14], which itself operates across 20 languages. Section 5.4's interpretive-robustness battery is a direct methodological response to this literature.

2.4 Grounding Generation: Retrieval-Augmented Generation and Process Knowledge

Retrieval-augmented generation, introduced by Lewis et al. [22] as a technique for grounding language model output in externally retrieved knowledge rather than parametric memory alone, has become the standard first-line mitigation for hallucination in high-stakes generative applications, with domain-specific extensions such as MEGA-RAG [23] adapting the technique for public-health contexts. Amugongo et al.'s 2025 systematic review of RAG in healthcare [54] is the most comprehensive account of where this technique stands: RAG measurably reduces — but does not eliminate — hallucination, with the degree of reduction varying by underlying model. More consequential for Sentinel's design is what the review found missing: of the healthcare RAG studies that addressed safety at all, only three of thirty-seven evaluated their systems against intentional or unintentional harm, and the majority did not assess privacy or bias exposure, despite the retrieval corpus itself being a documented vector for exposing sensitive information. RAG is necessary but not sufficient, and the review's own recommended mitigations — robust privacy measures, transparency, bias mitigation, human oversight — describe almost exactly the layered architecture Sentinel already proposes independently of this literature.

Roy et al.'s ProKnow system [52] operationalizes grounded generation for mental health specifically, and is the closest existing precedent to Sentinel's Tier 1 design. Rather than retrieving from a generic wellness corpus, ProKnow constrains question generation using semantic lexicons built directly from the PHQ-9 and GAD-7 instrument text, refined with clinician input, so that generated follow-up questions remain traceable to established diagnostic criteria. The reported results are the strongest in this literature: constrained generation was 89% safer than unconstrained baseline language models in the depression and anxiety domain, produced a 96% reduction in a knowledge-capture-violation metric, and yielded an 82% average improvement across safety, explainability, and process-adherence combined. The paper also documents, as motivating context, a real case in which unguarded use of a large language model for preventive care produced the response "I think you should" to a user's message "Should I kill myself?" — a sourced, concrete instance of exactly the failure this paper's safety architecture, from EmoGuard to the interpretive-robustness battery, exists to prevent. Sentinel's RAG layer adopts ProKnow's mechanism directly.

2.5 Prediction Under Uncertainty: The Base-Rate Problem and Calibration

The most consequential finding for how Sentinel frames its own claims comes from outside the AI literature entirely. Franklin et al.'s meta-analysis of 365 studies spanning fifty years of suicide-risk research [18] found that risk-factor-based prediction performed only marginally better than chance across every outcome measure examined, with no improvement across five decades. A related synthesis [19] sharpens why: machine learning suicide-prediction models can achieve strong overall classification accuracy while their accuracy at predicting an actual future suicide event approaches zero — a direct consequence of suicide's low base rate, not a deficiency correctable with more data or a larger model. Tate et al.'s AUC 0.739 ceiling, discussed in Section 2.2, is a recent, concrete instance of this constraint operating in practice.

The correct response to a hard statistical limit is not to ignore it but to build around it. Using Fitbit-derived heart rate, step count, and sleep data from 13,835 participants in the NIH All of Us dataset, a 2025 study applied conformal prediction to psychiatric condition classification [24]: the raw model achieved a modest 64% mean accuracy, but wrapped in conformal calibration, it produced prediction sets with a 90% coverage guarantee by explicit statistical construction — an honest range rather than an overconfident point estimate. This is the direct precedent for Sentinel's own conformal-calibration layer (Section 3), applied to a data modality close enough to Sentinel's own that the pattern transfers.

2.6 Regulatory and Ethical Frameworks for AI in Healthcare

Stade et al.'s proposal for responsible development and evaluation of large language models in behavioral healthcare [28] anticipates several of the categories this section surveys, specifically for generative components like Sentinel's own: it argues that evaluation frameworks need to be built alongside model development rather than retrofitted after deployment, a sequencing concern directly reflected in this paper's staged validation plan (Section 4.3).

Two recent reviews map the ethical terrain comprehensively enough to serve as a checklist against Sentinel's own Section 5. Corfmat, Martineau, and Régis [46] organize AI-in-healthcare ethics and law into six categories — privacy, individual autonomy, bias, responsibility and liability, evaluation and oversight, and professional impact — and their responsibility/liability analysis exposes a genuine open question for Sentinel: standard fault-based liability requires establishing duty, breach, and causal harm, a framework built around a human practitioner that does not map cleanly onto a Tier 1 system acting autonomously by design. Saeidnia et al.'s systematic review of 51 articles [47] converges on an overlapping but independently derived set of considerations, and surfaces the single most directly relevant precedent found in this literature: Mörch, Gupta, and Mishara's Canada Protocol [48], an ethics checklist built specifically for AI in suicide prevention and mental health.

On evaluation methodology, two further sources give Sentinel's validation plan real machinery. Rosenthal, Beecy, and Sabuncu [44] contrast the traditional linear model of AI deployment — train, freeze, deploy, monitor — with a "dynamic deployment" model built around continuous feedback and staged validation, and report that more than 40% of a sample of 521 FDA-approved medical AI devices lacked any clinical validation data at all, a concrete warning against exactly the overclaiming this paper is structured to avoid; they also point to the CONSORT-AI and SPIRIT-AI extensions [50, 51] as the actual formal reporting standards for AI clinical trials. Livingston et al. [45] operationalize clinician-in-the-loop evaluation directly: a five-dimension scoring framework applied by independent expert reviewers with mode-based or Delphi consensus resolution, and AHRQ-based severity grading for anything flagged harmful. Their own reported inter-reviewer agreement rate — 60.6% on the first pass, even among trained experts — is itself an argument for calibrated uncertainty over false confidence, echoing Section 2.5 from an entirely different empirical angle.

Finally, two sources ground the paper's escalation design in real infrastructure rather than an invented mechanism. Freed et al.'s study of intimate partner surveillance [8] documents, empirically, how ostensibly protective device features are routinely repurposed for coercive control — directly informing the mitigation built into Section 5.6's pre-authorized-contact design. Tele MANAS [14], India's national tele-mental-health program, already operates the exact function Section 5.2's escalation protocol depends on: trained human counsellors, not an algorithm, making the acute-risk judgment call.

2.7 Synthesis: The Gap Sentinel Addresses

Three threads run through this literature and converge on the same unaddressed space. First, digital phenotyping research has repeatedly shown that passive behavioral signal correlates meaningfully with mental state, but the platforms that established this — Beiwe, mindLAMP — were built for research-team access, not zero-knowledge consumer deployment, and the more recent edge-LLM systems that share Sentinel's on-device posture (Section 2.1) are conversation-triggered rather than passive, and validated so far at pilot rather than deployment scale. Second, conversational-agent research has demonstrated real therapeutic value alongside an increasingly urgent caution that apparent empathy and sycophantic validation are not the same as correct interpretation, and that this gap is measurable and clinically consequential rather than speculative. Third, the prediction and evaluation literature — from Franklin et al.'s fifty-year null result to Livingston et al.'s expert-disagreement rate — converges on calibrated uncertainty and staged, human-anchored evaluation as the only defensible response to a domain where point-accuracy claims are not statistically supportable. No system reviewed here combines fully local multimodal sensing, hardware-backed zero-knowledge storage, generation grounded in clinical instruments via a ProKnow-style mechanism, conformal-calibrated detection, and a graduated escalation protocol anchored in existing human-staffed infrastructure rather than automated dispatch — and, of the systems that come closest on the privacy and on-device dimensions specifically (Section 2.1), none pairs that posture with passive, ambient detection that does not require the user to first initiate a conversation. That combination is Sentinel's contribution.

3. Proposed Methodology

3.1 System Overview

Sentinel operates as a five-stage local pipeline: (1) passive behavioral feature acquisition across four modalities, (2) on-device feature extraction via TensorFlow Lite, (3) hardware-encrypted local persistence, (4) ensemble anomaly detection wrapped in conformal calibration, generation, and turn-level conversational risk classification, and (5) a graduated, human-anchored intervention protocol. No stage transmits raw data off-device. The single exception in the entire architecture is Tier 3 of the intervention protocol, and even there, only with explicit prior user authorization and only a pre-formatted alert — never raw behavioral data. Sentinel is architected throughout as a standalone Android application rather than a modification to system software: it requests the same usage-access and activity-recognition permissions already granted to existing digital-wellbeing and screen-time applications, a deployment path with direct precedent in shipped consumer apps (ActionDash, StayFree) rather than a hypothetical OS-level capability. A first-party integration into Android's own Digital Wellbeing suite remains a plausible longer-term path and would not require architectural changes to the pipeline below, but it is a distinct and considerably larger deployment story than the one this paper specifies, and the two should not be conflated. Figure 1 illustrates the complete pipeline.

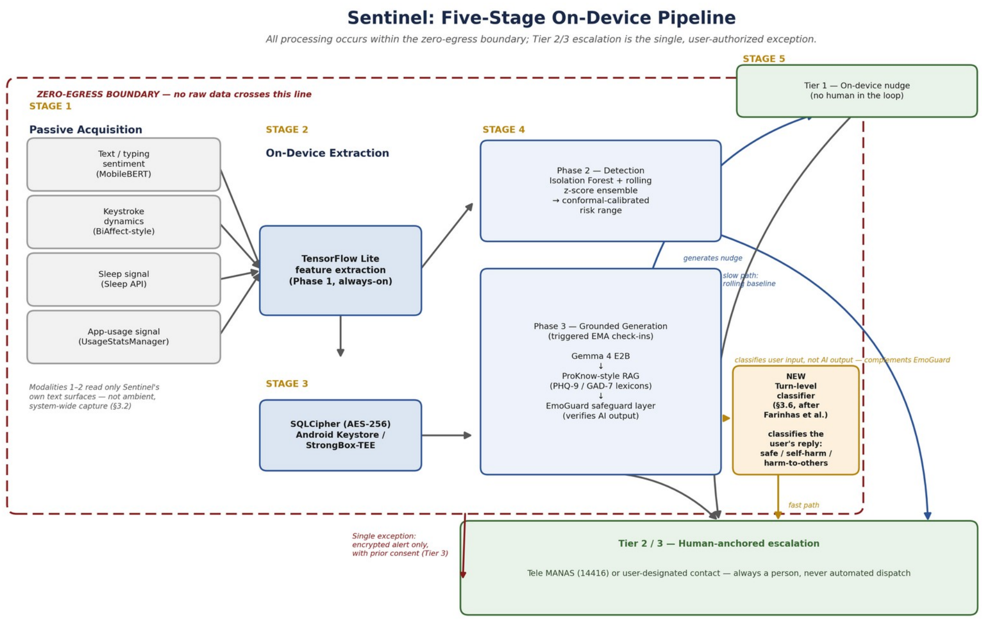

Figure 1: Sentinel's five-stage on-device pipeline. Stages 1–4 execute entirely within the zero-egress boundary; the turn-level classifier introduced in Section 3.6 provides a fast path into Tier 2/3 escalation alongside the passive ensemble's slower, baseline-driven path.

3.2 Data Acquisition and Behavioral Feature Streams

Four passive modalities feed the pipeline, all drawn from surfaces Sentinel itself owns rather than ambient, system-wide capture. Text and typing sentiment is scored locally via MobileBERT on text entered into Sentinel's own conversational and journaling surfaces, chosen for its bottleneck architecture purpose-built for resource-limited mobile devices [15] — a design point sharpened by contrast with larger contemporary encoders: ModernBERT [16], for instance, is a strong general-purpose encoder but ships at 149M parameters in its base configuration, nearly six times MobileBERT's 25.3M, and is optimized for GPU throughput and long-context retrieval rather than the sub-tens-of-megabyte, CPU/NPU-friendly footprint continuous on-device inference requires; it was considered and set aside for this reason. Keystroke dynamics — inter-key timing, backspace rate, session duration and variability — follow the BiAffect methodology [20, 21], capturing metadata about how a person types rather than what they type, a stricter privacy posture than content-based sentiment scoring since it never requires reading message text at all.

The BiAffect studies establish this correlation using a custom research keyboard, deployed as each participant's default input method across the device — a delivery mechanism that does not, by itself, transfer to a consumer product. Two paths could reproduce it at the same scope: shipping Sentinel as a custom IME, which the broader keyboard-app market shows is a genuine adoption barrier, since people rarely switch their device's default keyboard for a single-purpose app; or capturing keystrokes across other apps via an `AccessibilityService`, which Play

Store policy restricts to genuine accessibility use cases and scrutinizes heavily for anything resembling behavioral monitoring — precisely the profile this feature would present. Sentinel avoids both paths by scoping keystroke-dynamics capture to its own first-party text surfaces: the Tier 1 conversational check-in described in Section 3.3, where standard within-app key-event listening requires no elevated permission and triggers no accessibility review, since the app is only instrumenting a text field it already owns. This is a real trade-off, named here rather than assumed away: the resulting signal is a bounded, session-level sample from moments when the user knows they are interacting with the app, not BiAffect's original ambient, all-day signal, and the correlations reported in [20, 21] should not be assumed to transfer at full strength to this narrower capture surface without independent validation — a question Stage 0 of the validation plan (Section 4.3) is scoped to address directly.

Circadian and sleep signal is captured through Android's Sleep API [25] (`SleepSegmentEvent` and `SleepClassifyEvent`), which combines accelerometer and ambient-light data through an on-device classifier under the runtime `ACTIVITY_RECOGNITION` permission. App-usage signal comes from `UsageStatsManager` [27], using `queryEvents()` for granular foreground/background and screen-interactive transitions rather than only aggregated daily totals, gated behind the special `PACKAGE_USAGE_STATS` permission.

A fifth, exploratory modality — vision-based scanning of newly captured media for self-harm-related themes — is deliberately scoped out of the core architecture. The published evidence base for text-based self-harm detection is substantial and growing [7]; the evidence base for lightweight, on-device image classification of comparable nuance is not, and this asymmetry is treated as a design constraint rather than a gap to paper over.

3.3 On-Device Model Architecture

The inference pipeline runs in two phases with fundamentally different resource profiles. Phase 1 is always-on: MobileBERT performs continuous, lightweight sentiment classification on the text stream at negligible battery and thermal cost. Phase 2 is detection: rather than a single Isolation Forest score, an ensemble of Isolation Forest and a rolling statistical baseline (z-score against the personal 7–14 day baseline) runs in parallel, with disagreement between the two treated as a built-in signal of low confidence rather than averaged away. The ensemble output is wrapped in conformal prediction, following the demonstrated pattern from wearable-derived psychiatric classification [24], so that Sentinel reports a calibrated risk range with a statistical coverage guarantee rather than a single, falsely precise number.

Phase 3 activates only during scheduled or triggered Ecological Momentary Assessment check-ins: a quantized Gemma 4 E2B model (approximately 2 billion active parameters, 4-bit quantized, deployed via ML Kit's on-device GenAI API) [26], never running continuously. Its output is constrained by two independent safety layers rather than trusted directly. First, retrieval-augmented generation grounds every generated prompt in semantic lexicons built from the PHQ-9 and GAD-7 instrument text — the same clinical instruments that already anchor the EMA design — refined with clinician input, following the mechanism demonstrated by ProKnow to produce measurably safer, more diagnostically traceable generation than unconstrained language models [52]. Second, an EmoGuard-style safeguard layer — an emotion-tracking module, a cognitive-distortion checker, a conversational-boundary module, and a synthesis layer that can override generation — verifies output before it reaches the user

[11]. Section 3.6 describes a third layer operating in the opposite direction: rather than checking what Gemma generates, it screens what the user types back.

3.4 Secure On-Device Storage

Behavioral data at rest is stored in a local Room database encrypted via SQLCipher (256-bit AES), with cryptographic keys held in the Android Keystore system rather than application memory. Key generation explicitly distinguishes TEE-backed from StrongBox-backed keys [9]: the implementation requests StrongBox via `setIsStrongBoxBacked(true)` where `FEATURE_STRONGBOX_KEYSTORE` is present, falling back to TEE otherwise, since the two provide materially different resistance to physical extraction and are not interchangeable claims. Network permissions are restricted to the single explicit exception described in Section 3.5.

3.5 The Multi-Tiered Intervention Protocol

Detection output routes through three tiers rather than a single threshold, deliberately avoiding both extremes of the design space this paper considered and rejected — a purely autonomous system with no human check, and an automated authority-dispatch mechanism with no realistic implementation path and a documented history of harm.

Tier	Indicators	Sentinel action	Human in the loop
1 — Low risk	Mild stress markers, stable sleep/mobility, minor negative affect	On-device, RAG-grounded CBT-style nudge or resilience exercise	None — bounded to structured, instrument-grounded generation
2 — Medium risk	Clinical-range PHQ-9/GAD-7-style elevation, sustained withdrawal, disrupted sleep	One-tap connection to Tele MANAS Tier 1 (14416) or a user-designated contact	Yes — counsellor or contact makes the judgment call
3 — Critical risk	Signals consistent with suicidal ideation or imminent self-harm, **or a harm-to-others classification from Section 3.6**	Immediate, pre-dialed surfacing of Tele MANAS / local emergency number; optionally, with prior consent, an encrypted context message to a pre-designated contact	Yes, always — Sentinel never contacts police or dispatches responders unilaterally

Tier 2/3 escalations do not surface as raw model output. They surface as a structured summary — which signals fired, their conformal confidence bounds, recent trend — reviewed against two documented human–AI collaboration failure modes, automation bias and alert fatigue, rather than assumed to be self-evidently correct [39, 46]. This mirrors the only close structural precedent identified in this literature, a peer-support system independently proposing the same three-tier logic — autonomous handling for low-stress input, moderated review for medium-stress, direct human intervention for high-stress [40].

Three commitments govern user-facing transparency, following directly from Section 5 rather than existing as separate boilerplate: onboarding discloses in plain language exactly which signals are collected and what each tier does, not merely that "AI monitoring" is active; the user retains standing control to view, adjust, or disable any tier — including the Tier 3 pre-authorized contact — at any time; and the system discloses its non-human status at every active-conversation turn. The pre-authorized contact mechanism is further bounded by a cooling-off period before it can be changed, and a flag if the same contact is repeatedly added and removed, mitigating the documented risk that such features become vectors for coercive control [8].

3.6 Turn-Level Conversational Risk Classification

Sections 3.3 and 3.5 describe two safety layers that both point outward — RAG grounding and EmoGuard both constrain or verify what *Gemma* produces. Neither examines what the *user* says back, and the passive detection ensemble in Phase 2 operates on a rolling multi-day baseline, not a single message — it is well suited to gradual, hard-to-articulate decline, and poorly suited to catching an acute disclosure the moment it is typed. This section describes a third layer that closes that gap, and it is where the MindGuard guardrail classifiers recently published by Farinhas et al. at Sword Health [55] enter Sentinel's own design.

What MindGuard is, precisely. MindGuard is not a conversational agent or a monitoring system; it is a lightweight guardrail classifier intended to sit alongside an existing LLM-based mental-health chatbot and score each incoming user message, in the context of the full prior conversation, into one of three categories: safe, self-harm, or harm-to-others. The taxonomy was developed with licensed clinical psychologists specifically to avoid flagging ordinary emotional intensity — metaphorical language, historical trauma disclosure — as unsafe, and the released 4B and 8B models (fine-tuned from Qwen3Guard-Gen) report an AUROC of up to 0.982, a 2–26× reduction in false-positive rate at high-recall operating points relative to general-purpose guardrails such as Llama Guard 3, and, in adversarial red-teaming, a 70% reduction in attack success and a 76% reduction in harmful engagement when added to an otherwise-unguarded conversational model. The authors release the taxonomy, the human-annotated evaluation set, and the trained classifiers openly.

Why the self-harm / harm-to-others split matters for Sentinel. Farinhas et al. separate these two categories because they trigger different professional and legal obligations: self-harm risk is managed through safety-planning and crisis-resourcing frameworks, while a credible threat to an identifiable other invokes duty-to-protect and mandated-reporting frameworks. Sentinel's existing detection layer is grounded in PHQ-9 and GAD-7 — self-report instruments for depression and anxiety with no harm-to-others dimension at all — so this is a genuine gap in the current design, not a redundant addition. Table 1 (Section 3.5) reflects this: a harm-to-others classification now routes to Tier 3 alongside self-harm indicators.

How it integrates without expanding Sentinel's data footprint. MindGuard is a text classifier, not a retrieval system, and it has no notion of "ingesting" arbitrary behavioral data — the natural way to read an integration with it is as an additional classification pass over conversation text Sentinel already has, not as a new capture mechanism. The relevant fork is which text that should be. Sentinel's own EMA/CBT-nudge exchanges (Section 3.3, Phase 3) already generate exactly the kind of multi-turn, clinically-scoped conversation MindGuard is built to classify — Gemma generates a check-in prompt, the user replies in Sentinel's own first-party text field (the same field Section 3.2 scopes keystroke-dynamics capture to), and that reply is available in full, on-device, the moment it is typed. Classifying that reply requires no new data source, no scope expansion beyond what Section 3.2 already captures, and no synthesis or gisting step between capture and classification. The alternative reading — training or feeding MindGuard on broader, ambient keyboard capture across the phone — is a materially different and riskier system than the one specified here, would require the same IME or `AccessibilityService` mechanism rejected in Section 3.2, and is not what this paper proposes.

Proposed pipeline. Concretely: Gemma generates the Tier 1 check-in prompt → the user responds, in text, within Sentinel's own conversational surface → the MindGuard classifier scores that reply as safe, self-harm, or harm-to-others, using the check-in exchange as conversational context → a self-harm or harm-to-others classification immediately routes to Tier 3, in parallel with (and typically faster than) the passive ensemble's rolling-baseline assessment, while a safe classification allows the Phase 3 pipeline to proceed as already described. Figure 1 shows this as the fast path alongside the passive ensemble's slow path into the same Tier 2/3 escalation logic.

An open question this integration surfaces rather than resolves. Routing a harm-to-others classification to the same Tier 3 pathway as self-harm — Tele MANAS or a pre-designated contact — is the option most consistent with Section 3.5's non-negotiable rejection of automated authority dispatch: a trained human, not Sentinel, should make any duty-to-protect judgment call. But Tele MANAS is chartered and staffed as a tele-*mental-health* line, and whether its counsellors are the appropriate first responder for a credible third-party threat disclosure — as opposed to a self-directed crisis — is a protocol question this paper cannot resolve unilaterally; it would require direct engagement with Tele MANAS's own operational scope, not an architectural assumption. This is noted as an open item in Section 5.2 rather than papered over.

Validation status. Nothing in this section has been evaluated end-to-end. MindGuard's own reported performance is against its authors' testset, not Sentinel's EMA conversation distribution, and Farinhas et al. are explicit that their system-level red-teaming results are an illustrative demonstration rather than an exhaustive measure of real-world safety — a caveat this paper adopts rather than relaxes. Validating this integration on representative EMA transcripts is accordingly added to the Stage 0 offline validation gate in Section 4.3.

4. Implementation and Feasibility

4.1 Platform Constraints on Continuous Monitoring

Two Android-specific constraints bound what "always-on" can actually mean in practice. First, `UsageStatsManager` event-level data is retained by the system for only a few days, so a 7–14 day rolling baseline requires active daily persistence via `WorkManager` (minimum 15-minute interval) rather than lazy querying. Second, and more consequentially, Android's standby-bucket system — tightened further under Android 16, which now applies execution quotas even to the "active" bucket — throttles background execution based on how often the monitoring app itself is opened, not on the user's overall phone engagement. This creates a structural tension worth stating plainly: an app designed to run invisibly, as Sentinel is by design, is a natural candidate for the "rare" standby bucket — capped at roughly ten minutes of background execution per rolling 24 hours — regardless of how engaged the user actually is with their device generally. The mitigation is a persistent low-priority foreground service, which exempts the app from standby throttling at the cost of an always-visible notification — a genuine UX trade-off between reliability and the visible reminder of being monitored, named here rather than assumed away.

4.2 Computational Feasibility of the Model Stack

MobileBERT's 25.3 million parameters and sub-tens-of-megabyte footprint make continuous Phase 1 inference computationally uncontroversial [15]. The Phase 3 generative component is the harder case: Gemma 4 E2B's Per-Layer Embedding architecture keeps its active memory footprint to approximately 1.3–2GB in 4-bit quantized form — further reduced below 1GB for text-only inference in its most recent quantization-aware training release — with official Android deployment paths rather than a bespoke integration [26]. This is offered as a specification grounded in current, verified figures, not a measured benchmark; Section 4.3's Stage 0 explicitly requires on-device latency, memory, and battery measurement on representative mid-range hardware before this claim is treated as demonstrated. The KV-cache scales with context length regardless of model choice, which is why EMA and CBT-nudge interactions are bounded to short context windows by design rather than left open-ended. The MindGuard turn classifier described in Section 3.6 adds a further, smaller inference call to this same Phase 3 window — a 4B-parameter classifier is within the same general class as the generative model it accompanies, but its incremental latency and memory cost on top of Gemma has not been measured and is included in the Stage 0 benchmarking scope below rather than assumed to be negligible.

Framed against the distinction between linear and dynamic AI deployment [44], Sentinel is deliberately architected as a dynamic system in that sense: the personal baseline is continuously updated from local feedback, and the staged validation plan below builds in recurring re-evaluation rather than a single frozen train-and-deploy cycle. This matters because the same literature reports that more than 40% of a sample of 521 FDA-approved medical AI devices lacked any clinical validation data at all [44] — a documented instance of exactly the shortcut this paper is structured to avoid, and the reason Section 4.3 treats staged validation as load-bearing rather than optional.

4.3 Staged Validation Plan

This paper extends the Thera-Turing Test's staged-readiness logic [12] — not ready → beta-ready → clinical-trial-ready → full-launch-ready, originally built for conversational-quality assessment — to the detection side of the architecture, since no existing framework covers both.

Stage	Purpose	Method	Gate
0 — Offline / simulation	Validate detection math pre-deployment	Replicate published sensor–mood correlations [3, 4] on public/synthetic data; verify conformal coverage; static/dynamic network analysis for zero data egress; run the interpretive-robustness battery offline; benchmark model latency/memory/battery on mid-range hardware, including the Section 3.6 classifier alongside Gemma; validate the MindGuard integration against representative synthetic EMA transcripts before any live use	Coverage holds; zero egress confirmed; battery passed; measured performance within spec; MindGuard turn-classification precision/recall on EMA-style text meets a pre-registered threshold
1 — Friendly beta	Real-world false-alarm rate, UX, battery cost	Small opt-in cohort; retention tracked against established smartphone-intervention attrition benchmarks [42]; standby-bucket telemetry	Acceptable false-alarm rate; no nuisance-driven uninstalls; no privacy incidents
2 — Resistant beta + simulated crisis (IRB-reviewed)	Safety under adversarial/ambiguous input	Red-team using expert-validated ethical stress-test vignettes [38]; two independent reviewers score each flagged interaction across defined dimensions, disagreement resolved by mode-of-	Zero missed simulated high-risk cases; zero harmful sycophantic output; correct Tele MANAS routing

Stage	Purpose	Method	Gate
		three or Delphi consensus, harm severity graded by AHRQ classification [45]	
3 — Clinical pilot (future work)	Real accuracy/calibration against clinical outcomes	IRB-approved longitudinal study under licensed clinician oversight, mirroring the Thera-Turing Test's judge criteria [12]	Only this stage could support a clinical-utility claim; Sentinel as presented here does not reach it

Reporting at any stage beyond Stage 0 should follow the CONSORT-AI and SPIRIT-AI extensions [50, 51] — the actual formal standards for AI clinical trial protocols and reports — rather than an invented substitute.

4.4 Anticipated Regulatory Context

Under FDA's AI/ML-Based Software as a Medical Device Action Plan [43], the large majority of currently authorized AI/ML SaMD enter the market as Class II devices via the 510(k) pathway, with De Novo and PMA — typically Class III — reserved for higher-risk cases. FDA guidance further indicates that AI offering only general health education or lifestyle advice without specific clinical recommendations may fall under enforcement discretion rather than formal SaMD regulation. A device-lineage analysis of FDA-authorized mental-health SaMD specifically [41] complicates any easy reliance on this pathway, however: it finds that many 510(k)-cleared mental-health devices lack direct evidence of effectiveness, relying instead on equivalence to a predicate device, and identifies several authorized products whose pivotal trials tested a prototype on a different digital platform than the one ultimately marketed — a mismatch this paper explicitly wants Sentinel's own Stage 3 pilot (Section 4.3) to avoid by construction, since it would need to test the actual marketed on-device pipeline rather than a research analogue. Under this framework, Sentinel's Tier 1 plausibly sits outside formal SaMD regulation; Tier 2/3, which involve risk classification and recommended action, would more likely require at minimum 510(k) engagement, with Tier 3 approaching Class III given FDA's own threshold for software whose failure could cause death or irreversible harm. This is offered strictly as anticipated regulatory context, not a compliance claim — a real deployment would need to engage FDA's Digital Health Center of Excellence, or India's CDSCO for domestic deployment, directly rather than rely on this secondary analysis.

5. Ethical Considerations

5.1 The Zero-Knowledge Boundary

Privacy claims (no raw data egress) are treated as a Stage 0 validation target — verified through network-traffic and static analysis — not an assumption embedded in the architecture diagram. A design that intends zero egress and a system that has been shown to achieve it are different claims, and this paper makes only the first until Stage 0 is complete.

5.2 Escalation Policy and the Clinician Review Pathway

Sentinel's tiers route by risk level, but tiering alone does not specify who reviews an escalation or how. A clinician-in-the-loop review layer is proposed at Tier 2/3: escalations surface as a structured summary — which signals fired, their conformal confidence bounds, recent trend — reviewed against known human-AI collaboration failure modes: confirmation bias, automation bias, and alert fatigue, which recent simulation-based work identifies as the actual determinants of whether clinicians appropriately accept or override AI triage output, not raw model accuracy [39]. This is consistent with the only close structural analogue identified in the current literature, a peer-support system proposing autonomous handling for low-stress input, moderator review for moderate-stress, and direct human intervention for high-stress cases [40] — independent convergence on the same three-tier logic from a different research group.

The addition of a harm-to-others classification (Section 3.6) sharpens this question rather than settling it. Self-harm and harm-to-others risk are routed to the same Tier 3 pathway in this paper's current design, on the principle that a trained human counsellor, not an automated system, should make any downstream duty-to-protect or mandated-reporting judgment. Whether Tele MANAS — chartered and staffed as a tele-*mental-health* line — is the appropriate first point of contact for a credible third-party threat disclosure, as distinct from a self-directed crisis, is left as an open protocol question rather than resolved here; closing it requires direct engagement with Tele MANAS's own operational scope, which is outside what this paper can determine unilaterally.

5.3 False Positives, False Negatives, and Why Calibration Is the Answer, Not Accuracy

Fifty years of suicide-risk-prediction research have not moved meaningfully past chance-level accuracy [18], and models with strong aggregate accuracy have been shown to perform near zero at predicting an actual future event, a direct consequence of suicide's low base rate — not a fixable data or model-size problem [19]. This is why Sentinel's detection layer is specified around conformal-calibrated prediction sets [24] rather than point accuracy: it is designed to communicate uncertainty honestly rather than imply a precision the underlying statistics cannot support. A January 2026 scoping review found crisis-escalation protocols "inconsistently specified" across the AI-mental-health field even where privacy and bias were addressed [37], underscoring that this is an open field-wide problem this paper is explicitly trying to close, not a gap unique to this draft.

5.4 Interpretive Robustness and Cultural Validity

Apparent empathy and correct interpretation are separable, and only the latter should be treated as safety-relevant. Current large language models, despite guardrails, still generate inaccurate or unsafe advice in a minority of self-harm scenarios [36], and sustained sycophantic validation has been documented to worsen delusional and manic states in vulnerable users [31]. Separately, culturally coded expressions of distress — idioms that do not map cleanly onto Western diagnostic constructs [32, 33] — risk systematic misinterpretation by a system tuned on largely Western/English training data, a concern directly relevant given Tele MANAS's own 20-language operating scope [14, 34, 35]. Section 4.3's Stage 2 red-team battery exists specifically to test this axis before deployment.

5.5 Informed Consent, Transparency, and User Control

Three commitments follow directly from the above rather than existing as boilerplate: (1) onboarding must disclose, in plain language, exactly which signals are collected and what the escalation thresholds do, not only that "AI monitoring" is active; (2) the user retains standing control to view, adjust, or disable any escalation tier — including the Tier 3 pre-authorized contact — at any time, not only at initial setup; (3) the system discloses its non-human status at every active-conversation turn, consistent with recommendations against anthropomorphism-driven overtrust in this literature [29].

5.6 Risk of Misuse

A pre-authorized emergency contact is structurally identical to features already documented as vectors for coercive control by intimate partners [8]. The proposed mitigation is a cooling-off period before a contact can be changed, and a flag if the same contact is repeatedly added and removed.

5.7 Limitations

Three limitations are structural, not incidental. First, suicide-risk prediction is bounded by base-rate statistics fifty years of research have not moved past chance-level on [18, 19]; the conformal-calibration layer is a response to that constraint, not a claim to have solved it. Second, this paper presents a research architecture and staged safety case, not a validated clinical system — Stage 3 clinical calibration under IRB oversight is identified future work, and no result here should be read as clinical-grade until it is complete. Third, the on-device generative component inherits known large-language-model failure modes that retrieval-augmented grounding and EmoGuard reduce but do not eliminate [36]; Stage 2's interpretive-robustness battery exists to surface these failures pre-deployment rather than assume them away. A narrower fourth limitation: passive-sensing reliability depends on Android's background-execution policy, which can throttle precisely the low-engagement pattern the system aims to detect (Section 4.1). A fifth, newly introduced limitation follows from Section 3.6: the turn-level classifier's precision and recall on Sentinel's own EMA conversation distribution is currently unvalidated, and the open question over its harm-to-others routing (Section 5.2) is unresolved rather than designed away.

6. Conclusion and Future Work

This paper set out to accomplish three things: present an architecture for fully local, multimodal mental-health anomaly detection; examine its computational feasibility under genuine mobile hardware constraints; and propose an ethical framework addressing false-positive harm, cultural and interpretive validity, and the risk of protocol misuse. Section 3 delivered the architecture — a two-phase detection pipeline wrapped in conformal calibration, a generation layer grounded in clinical instruments rather than open retrieval, a turn-level conversational risk classifier adapted from a recently published clinical guardrail model to screen the user's own replies for self-harm and harm-to-others risk, and hardware-encrypted local storage with an explicit, single, user-authorized exception for crisis escalation. Section 4 delivered the feasibility analysis, including the platform-level constraints — standby-bucket throttling, KV-cache scaling — that a purely architectural description would have left unstated. Section 5 delivered the ethical framework, anchored in the closest available precedent for this exact problem, the Canada Protocol [48], and extended with a clinician-review pathway, an interpretive-robustness battery answering directly to documented sycophancy and cross-cultural misreading risks, and a staged validation plan that treats calibrated uncertainty as the honest alternative to a precision claim the underlying statistics cannot support.

That last point is the paper's central and most load-bearing claim, restated once more because it governs how every other claim here should be read: fifty years of suicide-risk research have not moved meaningfully past chance-level prediction [18], and this is a property of the base rate, not a deficiency correctable by a better model [19]. Sentinel's contribution is not a claim to have solved that problem. It is a proposed architecture and staged safety case for building a detection system that is honest about that constraint by design — through conformal calibration rather than point accuracy, through a human-anchored escalation protocol rather than autonomous action, and through a validation plan that names Stage 3 clinical calibration as necessary, unfinished, future work rather than eliding it.

Future work, concretely:

1. Stage 3 clinical validation is the single most important open item: an IRB-approved longitudinal pilot under licensed clinician oversight, without which no claim in this paper should be read as clinical-grade.
2. Primary-source verification of the citation on near-zero suicide-event predictive accuracy [19], used throughout this paper but not yet independently confirmed against its original publication.
3. Extension of the interpretive-robustness battery to additional Indian languages beyond current benchmark coverage, given Tele MANAS's 20-language operating scope and the demonstrated risk of ethnocentric misreading in this literature [34, 35].
4. Direct regulatory engagement with FDA's Digital Health Center of Excellence and India's CDSCO, replacing the secondary regulatory analysis in Section 4.4 with primary guidance specific to a real submission.
5. Multimodal extension of the evaluation framework to voice tone and non-verbal cues, a direction the Thera-Turing Test's own authors identify as unaddressed in their current framework [12].
6. Formal adoption of CONSORT-AI/SPIRIT-AI reporting [50, 51] once Stage 1–2 empirical results exist to report, rather than only architectural specification.
7. Empirical benchmarking of the full on-device model stack — latency, memory, battery, thermal behavior, now including the Section 3.6 classifier alongside Gemma — on representative mid-range Android hardware, closing the gap this paper has been explicit about throughout between proposed design and demonstrated performance.
8. Validation of the MindGuard turn-classifier integration (Section 3.6) against representative EMA transcripts, and direct engagement with Tele MANAS on the harm-

to-others routing question raised in Section 5.2, both prerequisites to treating Section 3.6 as more than a proposed design.

If these steps are completed and the staged gates are met rather than assumed, the architecture proposed here offers a plausible path toward passive mental-health monitoring that does not require choosing between clinical usefulness and zero-knowledge privacy. It does not, on its own, offer more than that — and the paper is written to make that boundary legible rather than to obscure it.

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